# Functions

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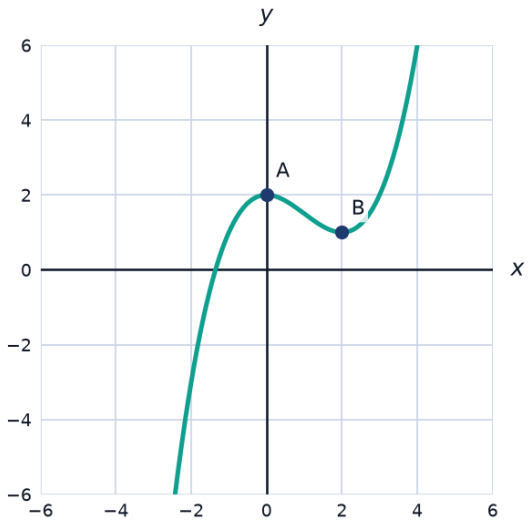
## Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

**1** [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Prove by direct deductive reasoning that the difference between the squares of any two odd integers is always divisible by 8.

**2** [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

The sum of the first $n$ terms of an arithmetic series is given by $S_n = 4n^2 - 3n$. (a) Find the first term $u_1$ and the second term $u_2$. (b) Find the common difference $d$. (c) Find an explicit expression for the $n$th term $u_n$, and calculate $u_{25}$.




**3** [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

In triangle $ABC$, side $a = 7$ cm, side $b = 9$ cm, and angle $A = 42^\circ$. (a) Show that two distinct triangles can be formed with these given measurements. (b) Calculate both possible values for angle $B$ and the corresponding values of side $c$.

**4** [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Three mobile antennas are located at coordinates $A(0, 0)$, $B(6, 0)$, and $C(2, 4)$. (a) Find the equation of the perpendicular bisector of segment $AB$. (b) Find the equation of the perpendicular bisector of segment $AC$. (c) Find the coordinates of the Voronoi vertex (circumcentre) equidistant from all three antennas, and calculate the radius of the circumcircle.

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**5** [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Six wines were evaluated by two independent sommeliers, $X$ and $Y$, giving the following ratings: Wine: A, B, C, D, E, F Judge X: 88, 72, 95, 60, 81, 75 Judge Y: 84, 78, 91, 65, 85, 70 (a) Construct a table of ranks for both judges. (b) Calculate Spearman's rank correlation coefficient $r_s$. (c) State what this coefficient indicates about the judges' agreement.

**6** [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A small business borrows USD 40,000 to renovate its premises. The loan carries an annual nominal interest rate of 7.2% compounded monthly, with equal monthly repayments made at the end of each month over 6 years. (a) Calculate the monthly repayment amount. (b) Calculate the total amount repaid and the total interest charged on the loan. (c) After making payments for 3 full years, determine the outstanding loan balance.

**7** [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A particle moves along a straight line such that its velocity in $\text{m/s}$ at time $t$ seconds is given by $v(t) = 6t - 2t^2$ for $0 \le t \le 4$. (a) Find the acceleration of the particle at $t = 2$ s. (b) Determine the net displacement of the particle between $t = 0$ and $t = 4$. (c) Calculate the total distance travelled by the particle over the interval $[0, 4]$.

## Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

**8** [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A curve has equation $f(x) = \frac{2x^2}{x^2 + k}$ for some positive constant $k > 0$. (a) Write down the horizontal asymptote of $y = f(x)$. (b) Find $f'(x)$ in terms of $k$. (c) Given that the gradient of the tangent to the curve at $x = 1$ is equal to $\frac{1}{2}$, find all exact possible values of $k$.

**9** [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A closed cylindrical metal container with circular base of radius $r$ cm and height $h$ cm has a fixed volume of $250\pi$ cm$^3$. (a) Express $h$ in terms of $r$. (b) Show that the total surface area is $A(r) = 2\pi r^2 + \frac{500\pi}{r}$. (c) Find the radius $r$ that minimizes the surface area, and show that for this optimal cylinder, the height equals the diameter ($h = 2r$).

**10** [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A continuous random variable $X$ is normally distributed with mean $\mu$ and standard deviation $\sigma$. Given that $P(X < 45) = 0.12$ and $P(X > 85) = 0.08$, find the value of $\mu$ and $\sigma$ correct to 3 significant figures.

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**11** [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Consider the function $f(x) = x^2e^{-x}$. (a) Find $f'(x)$ and express it in factored form. (b) Hence, find the coordinates of all stationary points of $f$. (c) Use the second derivative test to classify each stationary point as a local maximum or local minimum.

**12** [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Find the exact area of the finite region enclosed between the parabola $y = 3x - x^2$ and the line $y = x - 3$.

**13** [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

In triangle PQR, angle PQR = 90 degrees. S is a point on PR such that QS is perpendicular to PR. The length of PQ is 15 cm and the length of QR is 20 cm. Let the length of PS be x cm and the length of SR be y cm. Find the value of $x$.

**14** [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

The lifespan $X$ in hours of an electronic component is normally distributed with mean $\mu = 1200$ hours and standard deviation $\sigma = 150$ hours. (a) Find the probability that a randomly selected component lasts more than 1350 hours. (b) In a batch of 500 components, estimate how many will last between 1000 and 1400 hours. (c) The manufacturer guarantees the top 10% longest-lasting components. Determine the minimum lifespan required to receive this guarantee.

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## Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

**15** [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 1.01 Number & Algebra

In a factory, 60% of components are manufactured by Machine $A$ and 40% by Machine $B$. It is known that 3% of components produced by Machine $A$ are defective, whereas 5% of components produced by Machine $B$ are defective. (a) Find the probability that a randomly chosen component is defective. (b) Given that a randomly chosen component is defective, find the probability that it was produced by Machine $B$. (c) Determine whether the events "component produced by Machine B" and "component is defective" are independent.

**16** [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 3.01 Geometry & Trigonometry

In a 2D Cartesian plane, points $A, B, C$ have coordinates $A(2, -1)$, $B(5, 3)$, and $C(-1, k)$. (a) Find the vectors $\vec{AB}$ and $\vec{AC}$ in component form. (b) Find the value of $k$ for which $\vec{AB}$ is perpendicular to $\vec{AC}$. (c) For $k = 3$, calculate the exact cosine of the angle between $\vec{AB}$ and $\vec{AC}$.

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**17** [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 4.01 Statistics & Probability

Consider the binomial expansion of $\left(x^2 - \frac{2}{x}\right)^9$. (a) Write down the general term in terms of $r$. (b) Find the coefficient of $x^3$. (c) Find the term independent of $x$.

**18** [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 5.01 Calculus

Practice 4.1: A multiple-choice examination consists of 20 independent questions, each having 4 options of which only one is correct. A student guesses randomly on all questions. Let $X$ be the number of correct answers. (a) State the distribution of $X$ including its parameters. (b) Find the probability that the student passes by scoring at least 10 correct answers. (c) Calculate the mean and standard deviation of $X$.

**19** [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 6.01 AI SL — Specific Topics

Practice 2.1: The function $f(x) = 3 - \ln(2x + 1)$ is defined for $x > -0.5$. Find an expression for $f^{-1}(x)$, state its domain and range, and write down the coordinates of the intercept of $y = f^{-1}(x)$ with the $y$-axis.

**20** [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 1.01 Number & Algebra

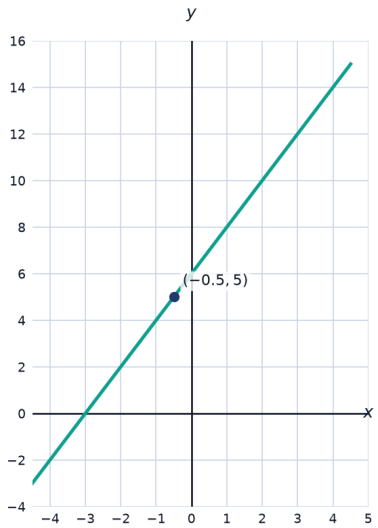
Practice 4.2: Bivariate data for study hours $x$ and exam mark $y$ across 10 students yield $\bar{x} = 14.5$, $\bar{y} = 72.0$, and Pearson's correlation coefficient $r = 0.88$. The regression line of $y$ on $x$ has equation $y = 2.4x + c$. (a) Find the value of $c$. (b) Interpret the slope $2.4$ in context. (c) Estimate the exam mark of a student who studies 18 hours.

## Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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**21** [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 2x + 6$ in cubic functions, evaluate $f(2)$.




**a.** 10

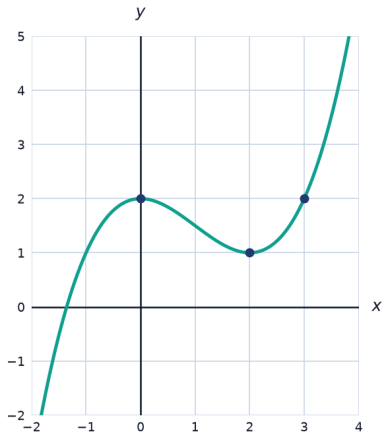
**b.** 12

**c.** 4

**d.** 14

**22** [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 3x + 7$ in power functions, evaluate $f(2)$.




**a.** 13

**b.** 16

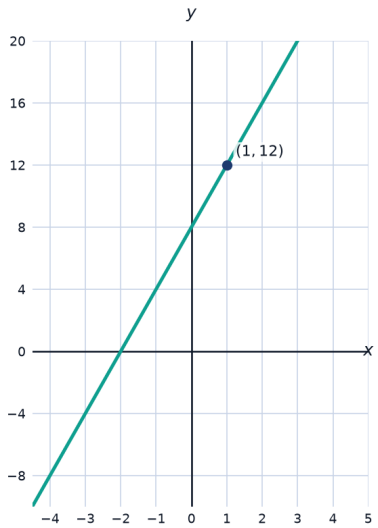
**c.** 6

**d.** 17

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**23** [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 4x + 8$ in direct variation, evaluate $f(2)$.




**a.** 16

**b.** 20

**c.** 8

**d.** 20

**24** [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for cubic functions consists of the four values $\{11, 15, 19, 25\}$. What is the arithmetic mean?

**a.** 15.5

**b.** 16.5

**c.** 17.5

**d.** 19.5

**25** [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for power functions consists of the four values $\{12, 16, 20, 26\}$. What is the arithmetic mean?

**a.** 16.5

**b.** 17.5

**c.** 18.5

**d.** 20.5

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**26** [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

A data set recorded for hyperbolas and inverse variation consists of the four values $\{14, 18, 22, 28\}$. What is the arithmetic mean?

**a.** 18.5

**b.** 19.5

**c.** 20.5

**d.** 22.5

**27** [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

For the model $f(x) = 5x + 9$ in hyperbolas and inverse variation, evaluate $f(4)$.

**a.** 29

**b.** 34

**c.** 20

**d.** 33

**28** [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

A data set recorded for transformations of hyperbolas consists of the four values $\{15, 19, 23, 29\}$. What is the arithmetic mean?

**a.** 19.5

**b.** 20.5

**c.** 21.5

**d.** 23.5

## Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

**29** [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

For the model $f(x) = 6x + 3$ in transformations of hyperbolas, evaluate $f(4)$.

**a.** 27

**b.** 33

**c.** 24

**d.** 31

**30** [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

For the model $f(x) = 2x + 4$ in graphing square root functions, evaluate $f(3)$.

**a.** 10

**b.** 12

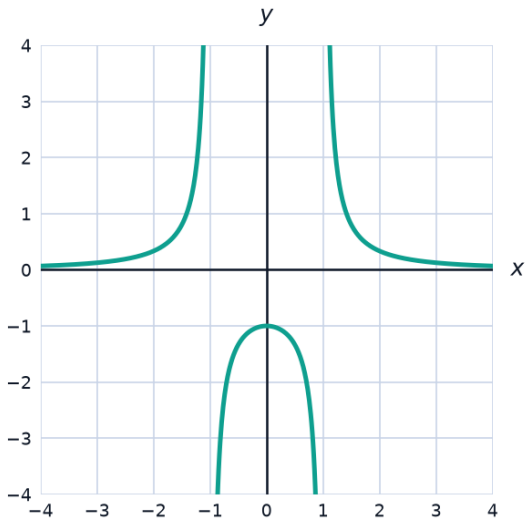
**c.** 6

**d.** 14

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**31** [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Consider the function $f(x) = \frac{1}{x-1}$ and the function $g(x) = x^2$. Let $h(x) = f(g(x))$. The graph of $y = h(x)$ is shown below. Determine the equation of the vertical asymptote(s) of $y = h(x)$.




**a.** x = 0

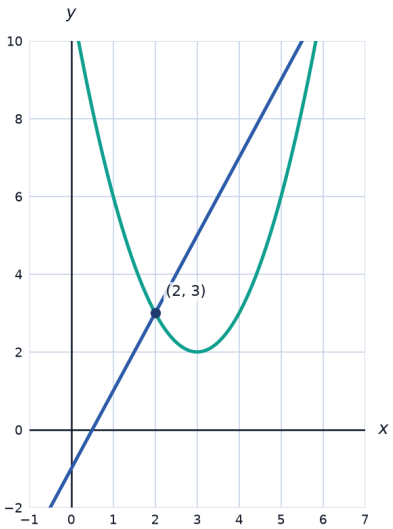
**b.** x = 1

**c.** x = -1

**d.** x = 0 and x = 1

**32** [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Consider the functions $f(x) = x^2 - 6x + 11$ and $g(x) = 2x - 1$. The graphs of these functions intersect at two points. One of the intersection points is $(2, 3)$. Find the x-coordinate of the other intersection point.




**a.** 3

**b.** 4

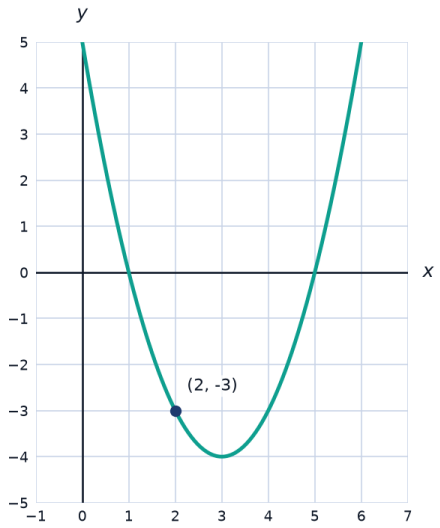
**c.** 5

**d.** 6

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**33** **[Intermediate]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

The graph of the quadratic function $f(x) = x^2 - 6x + k$ is shown below. The graph passes through the point $(2, -3)$. Determine the value of $k$.




**a.** $k = 5$

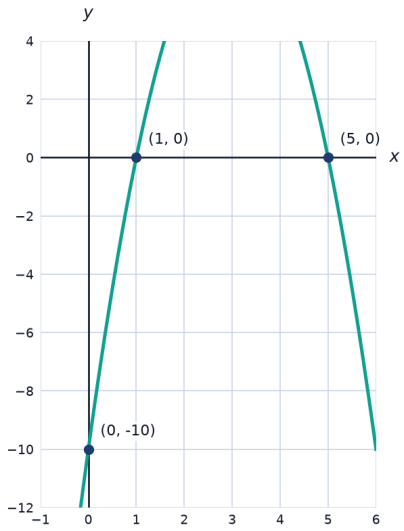
**b.** $k = -5$

**c.** $k = 1$

**d.** $k = -1$

**34** **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

The graph of the quadratic function $f(x) = ax^2 + bx + c$ is shown below. The graph passes through the points $(1, 0)$, $(5, 0)$, and $(0, -10)$. Find the value of $a$.




**a.** $-2$

**b.** $-1$

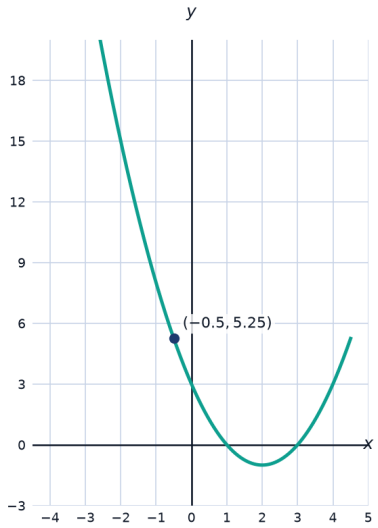
**c.** $1$

**d.** $2$

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**35** **[Intermediate]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)**

Consider the function $f(x) = x^2 - 4x + 3$. The graph of $y = f(x)$ is shown below. The function intersects the x-axis at two points. Find the coordinates of the point with the smaller x-value.




**a.** (1, 0)

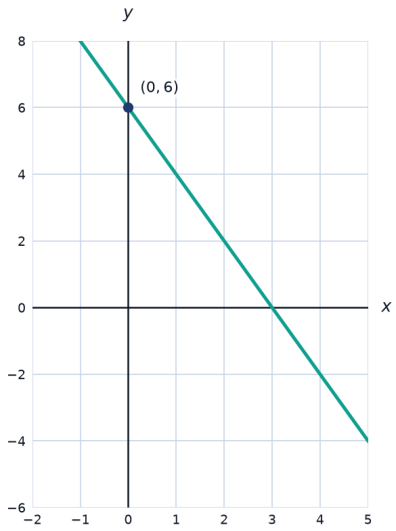
**b.** (3, 0)

**c.** (0, 3)

**d.** (0, 1)

**36** **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)**

Consider the function $f(x) = -2x + 6$. The graph of $y = f(x)$ is shown below. Determine the value of the $y$-intercept.




**a.** 6

**b.** -3

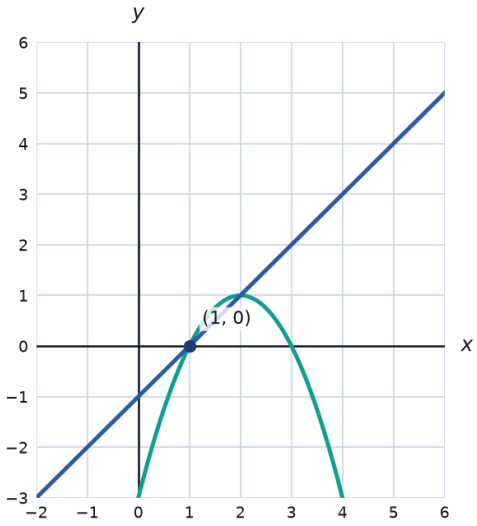
**c.** 3

**d.** -6

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**37** [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

The graph of the function $f(x) = -x^2 + 4x - 3$ is shown below. What are the coordinates of the y-intercept?




**a.** $(0, -3)$

**b.** $(0, 3)$

**c.** $(-1, 0)$

**d.** $(3, 0)$

## Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

**38** [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 5.01 Calculus

Consider the two functions $f(x) = ax^2 + bx + c$ and $g(x) = dx + e$. The graph of $f(x)$ is a parabola with its vertex at $(3, -1)$ and passes through the point $(1, 3)$. The graph of $g(x)$ is a straight line passing through the points $(2, 5)$ and $(6, -3)$. If the graphs of $f(x)$ and $g(x)$ intersect at two points, and the x-coordinate of one of these intersection points is $p$, find the value of $\frac{a+d}{p}$ given that $p$ is an integer.

**a.** 3

**b.** 2

**c.** -2

**d.** -3

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**39** [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 6.01 AI SL — Specific Topics

Consider the functions $f(x) = |x - a|$ and $g(x) = b$, where $a$ and $b$ are constants. The graph of $y = f(x)$ is a V-shaped graph with its vertex at $(a, 0)$. The graph of $y = g(x)$ is a horizontal line. If the graphs of $y = f(x)$ and $y = g(x)$ intersect at exactly one point, what is the relationship between $a$ and $b$ for the graphs to intersect at a single point?

**a.** a = b

**b.** a = -b

**c.** b = 0

**d.** a = 0

**40** [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 1.01 Number & Algebra

Consider the function $f(x) = 2x - 3$. Find the value of $f(3)$.

**a.** 3

**b.** 6

**c.** 9

**d.** 0

**41** [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 3.01 Geometry & Trigonometry

The point $(0, -3)$ lies on the graph of $y = f(x)$. Find the corresponding point on the graph of $y = f(x - 1) - 4$.

**a.** $(1, -7)$

**b.** $(-4, -2)$

**c.** $(1, 1)$

**d.** $(-1, -7)$

**42** [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 4.01 Statistics & Probability

Solve $2x^2 - 22x + 56 = 0$. Find the larger root.

**a.** 4

**b.** 7

**c.** 2

**d.** 3

**43** [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 5.01 Calculus

Let $f(x) = 3x^2 - 7x$. Find the equation of the axis of symmetry.

**a.** $x = -\frac{7}{3}$

**b.** $x = \frac{7}{3}$

**c.** $x = -\frac{7}{6}$

**d.** $x = \frac{7}{6}$

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**44** [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 6.01 AI SL — Specific Topics

Convert $210^\circ$ to radians, giving your answer as an exact multiple of $\pi$.

**a.** $\frac{5\pi}{3}$

**b.** $\frac{\pi}{2}$

**c.** $\frac{7\pi}{6}$

**d.** $\frac{7\pi}{4}$

**45** [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 1.01 Number & Algebra

Let $f(x) = 2x^2 + 6x - 5$. Find the y-intercept of the graph of $f$.

**a.** -15

**b.** -10

**c.** -5

**d.** 5

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